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| **Fiscal Year:** | FY 2009 | **Task Last Updated:** | FY 04/19/2009 |
| **PI Name:** | Jin, J.-P. M.D., Ph.D. | | |
| **Project Title:** | Adaptation Of Myocardial Contractile Proteins In Microgravity | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Developmental Biology<br>(2) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | jpjin@northwestern.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 847.570.1960 |
| **Organization Name:** | Evanston Northwestern Healthcare Research Institute | | |
| **PI Address 1:** | 2650 Ridge Ave | | |
| **PI Address 2:** | Molecular Cardiology Division | | |
| **PI Web Page:** | | | |
| **City:** | Evanston | **State:** | IL |
| **Zip Code:** | 60201-1718 | **Congressional District:** | 9 |
| **Comments:** | Former affiliation: Case Western Reserve University; moved in 2004. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology<br>01-OBPR-06 |
| **Start Date:** | 07/01/2004 | **End Date:** | 01/31/2009 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received extension to 1/31/2009 per A. Chu/ARC (6/2008)<br>NOTE: NCE to 6/30/2008 per A. Chu (4/07)<br>NOTE: NCE to 6/30/07 per A. Chu (7/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNA04CK62G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), a key protein in the Ca(2+)-regulatory system of cardiac muscle. This posttranslational regulation of cTnI selectively removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA) (Yu, Zhang, Jin, JBC 276:15753–60, 2001). Beta-adrenergic-regulated PKA phosphorylation of cTnI affects the sensitivity of cardiac muscle to Ca(2+). This research project tested the role of this up-regulation of N-terminal truncated cTnI in compensating for the reduced cardiac function in microgravity. The goal of this ground-based research was to identify a potential target for the prevention and treatment of cardiovascular deconditioning during and after space flight.<br>By investigating the structure-function relationship of cTnI and testing a transgenic mouse model over-expressing the N-terminal truncated cTnI in cardiac muscle, our experimental results have demonstrated that the selective deletion of the N-terminal extension of cTnI is not a destruction of the cardiac myofilament but provides a novel posttranslational regulation of cardiac function. The removal of the entire N-terminal domain of cTnI mimics the effect of PKA phosphorylation of Ser23/Ser24 in the N-terminal region of intact cTnI on enhancing myocardial relaxation. This proteolytic modification selectively utilizing a downstream mechanism of the beta-adrenergic signaling pathway to specifically enhance diastolic function of the heart while avoiding unwanted side effects. The enhanced ventricular relaxation due to up-regulation of the N-terminal truncated cTnI is plausible by compensating for the decreased circulating blood volume in long term space flight.<br><br>Strongly supporting this novel finding, we recently showed that a mouse model with beta-adrenergic deficiency is accompanied by up-regulated N-terminal truncation of cTnI. Double transgenic mice over-expressing the N-terminal truncated cTnI in the beta adrenergic deficient heart partially corrected the heart failure phenotype of beta-adrenergic deficiency (Feng, Chen, Weinstein, Jin, JBC 283:33384-93, 2008).<br><br>In conclusion, this highly rewarding research project has provided new information for understanding cardiac adaptation in microgravity and identified a potential target for enhancing cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our results contributed valuable new information to forwarding space life science as well as enhancing human life quality on Earth. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our discovery that the N-terminal modification of cardiac troponin I enhances cardiac relaxation has lead to an approach to the development of new drugs for the treatment of heart failure that is the most important death cause for cardiovascular diseases. |
| **Task Progress:** | This project was funded with a three-year budget originally from 12/02 to 11/05. The PI transferred this project from Case Western Reserve University to Evanston Northwestern Health care in 2004 and was granted no-cost-extensions until 01/09 to complete the proposed research work.<br>By investigating the structure-function relationship of cTnI and testing a transgenic mouse model over-expressing the N-terminal truncated cTnI in cardiac muscle, our experimental results have demonstrated that the selective deletion of the N-terminal extension of cTnI is not a destruction of the cardiac myofilament but provides a novel posttranslational regulation of cardiac function. The removal of the entire N-terminal domain of cTnI mimics the effect of PKA phosphorylation of Ser23/Ser24 in the N-terminal region of intact cTnI on enhancing myocardial relaxation. This proteolytic modification selectively utilizing a downstream mechanism of the beta-adrenergic signaling pathway to specifically enhance diastolic function of the heart while avoiding unwanted side effects. The enhanced ventricular relaxation due to up-regulation of the N-terminal truncated cTnI is plausible by compensating for the decreased circulating blood volume in long term space flight.<br><br>Strongly supporting this novel finding, we recently showed that a mouse model with beta-adrenergic deficiency is accompanied by up-regulated N-terminal truncation of cTnI. Double transgenic mice over-expressing the N-terminal truncated cTnI in the beta adrenergic deficient heart partially corrected the heart failure phenotype of beta-adrenergic deficiency (Feng, Chen, Weinstein, Jin, JBC 283:33384-93, 2008).<br><br>In conclusion, this highly rewarding research project has provided new information for understanding cardiac adaptation in microgravity and identified a potential target for enhancing cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our results contributed valuable new information to forwarding space life science as well as enhancing human life quality on Earth.<br><br>A) Five major research papers have been published:<br><br>68. Biesiadecki, B.J., Schneider, K.L., Yu, Z.-B., Chong, S.M., and Jin, J.-P. (2004) An Arg111Cys polymorphism in wild turkey cardiac troponin I accompanying the dilated cardiomyopathy-related abnormal splicing variant of cardiac troponin T with potentially compensatory effects. J. Biol. Chem. 279:13825-13832.<br><br>71. Barbato, J.C., Huang, Q.-Q., Hossain, M.M., Bond, M., and Jin, J.-P. (2005) Proteolytic N-Terminal Truncation of Cardiac Troponin I Enhances Ventricular Diastolic Function. J. Biol. Chem. 280:6602-6609.<br><br>81. Yu, Z.-B., Gao, F., Feng, H., and Jin, J.-P. (2006) Differential regulation of myofilament protein isoforms underlying the contractility changes in skeletal muscle unloading. Am. J. Physiol.:Cell Physiol., 292:C1192-203.<br><br>83. Yu, Z.-B., and Jin, J.-P. (2007) Removing the regulatory N-terminal domain of cardiac troponin I diminishes incompatibility during bacterial expression. Archives Biochem. Biophys. 461:138-45.<br><br>94. Chong, S.M., and Jin, J.-P. (2009) To investigate protein evolution by detecting suppressed epitope structures. J. Mol. Evol. in press<br><br>B) A comprehensive invited review has been published: 7. Jin, J.-P., Zhang, Z., and Bautista, J.A. (2008) Isoform diversity, regulation and functional adaptations of troponin and calponin. Crit. Rev. in Eukar Gene Expr. 18:93-124.<br><br>C) A US patent application claiming the N-terminal region of cTnI as a potential target for the detection of myocardial adaptation to stress conditions and for the treatment of cardiac diastolic dysfunction has been filed (reported in the 2007-2008 progress report). |

| Bibliography Type: | Description: (Last Updated: 04/20/2009) |
|---|---|
| Articles in Peer-reviewed Journals | Chong SM, Jin JP. "To investigate protein evolution by detecting suppressed epitope structures." J Mol Evol. 2009 Apr 14. [Epub ahead of print] PMID: 19365646 , Apr-2009 |
| Articles in Peer-reviewed Journals | McConnell BK, Popovic Z, Mal N, Lee K, Bautista J, Forudi F, Schwartzman R, Jin JP, Penn M, Bond M. "Disruption of protein kinase A interaction with A-kinase-anchoring proteins in the heart in vivo: effects on cardiac contractility, protein kinase A phosphorylation, and troponin I proteolysis." J Biol Chem. 2009 Jan 16;284(3):1583-92. Epub 2008 Oct 21. PMID: 18945669 , Jan-2009 |
| Patents | 11/311,472, patent application. US patent application serial # 11/311,472; most recent revision was submitted on 03/16/2009. Mar-2009 Jin J-P. "Methods for treating cardiac disease by modifying an N-terminal domain of troponin I." |